

which often is priceless. Conventional software programs cannot carry out the Herculean task of processing such data. AI is one of the few technologies that can handle this information overload.

We also need to distinguish between fact and fiction as far as the power of AI is concerned. I remember a talk by an expert in the field of AI a few years back. He talked about an AI based image recognition system that was able to classify the images of Siberian Huskies (a breed of dogs) and Siberian snow wolves with absolute or near absolute accuracy. Search the Internet and you will see how similar the two animals look. Had the system been so accurate, it would have been considered a triumph of AI. Sadly, this was not the case. The image recognition system was only classifying the background images of the two animals. Images of Siberian Huskies (since they are domestic animals) almost always had some rectangular or round objects in the background, whereas the images of Siberian snow wolves (being wild animals located in Siberia) had snow in the background. Such examples have led to the need for AI with some guarantee for accuracy in recent years.

Indeed, AI has shown some of its true power in recent years. A simple example is the suggestions we get from a lot of websites like YouTube, Amazon, etc. Many times, I have been astonished by the suggestions I have received as it almost felt as if AI was able to read my mind. Whether such suggestions are good or bad for us in the long run is a hot topic of debate. Then there is the critical question, “Is AI good or bad?” I believe that a ‘Terminator’ movie sort of future, where machines attack humans deliberately is far, far away in the future. However, the word ‘deliberately’ in the previous sentence is very important. AI based systems at present could malfunction and accidentally hurt humans. However, many systems that claim the powers of AI are conventional software programs with a large number of ‘if’ and ‘for’ statements with no magic of AI in it. Thus, it is safe to say that we are yet to see the real power of AI in our daily lives. But whether that impact will be good (like curing cancer) or bad (deepfake videos of world leaders leading to riots and war) is yet to be seen. On a personal level, I believe AI is a boon and will drastically improve the quality of life of the coming generations.

What is AI?

So, before we proceed any further, let us try to understand how AI, machine learning, deep learning, data science, etc, are related yet distinct from each other. Very often these terms are used (erroneously) as synonyms. First, let us consider a Venn diagram that represents the relationship between AI, machine learning, deep learning and data science (Figure 1). Please keep in mind that this is not the only such Venn diagram. Indeed, it is very plausible that you may find other Venn diagrams showing different relationships between the four different entities shown in Figure 1. However, in my opinion, Figure 1 is the most authentic diagram that captures the interrelationship between the different fields in question to the maximum extent.

First of all, let me make a disclaimer. Many of the definitions of the terms involved in this first article in this series on AI may not be mathematically the most accurate. I believe that formally defining every term with utmost precision at this level of our discussion is counterproductive and a wastage of time. However, in the subsequent articles in this series we will revisit these terms and formally define them. At this point in time of our discussion, consider AI as a set of programs that can mimic human intelligence to some extent. But what do I mean by human intelligence?

Imagine your AI program is a one-year old baby. As usual, this baby will learn his/her mother tongue simply by listening to the people speaking around him/her. He/she will soon learn to identify shapes, colours, objects, etc, without any difficulty at all. Further, he/she will be able to respond to the emotions of people around him. For example, any 3-year-old baby will know how to sweet talk his/her parents into giving him/her all the chocolates and lollipops he/she wants. Similarly, an AI program too will be able to sense and adapt to its surroundings, just

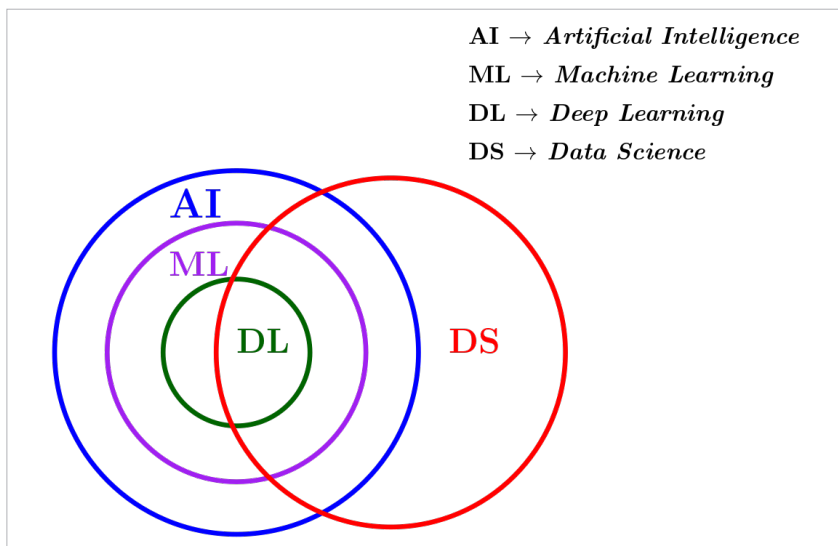


Figure 1: The AI hierarchy and data science